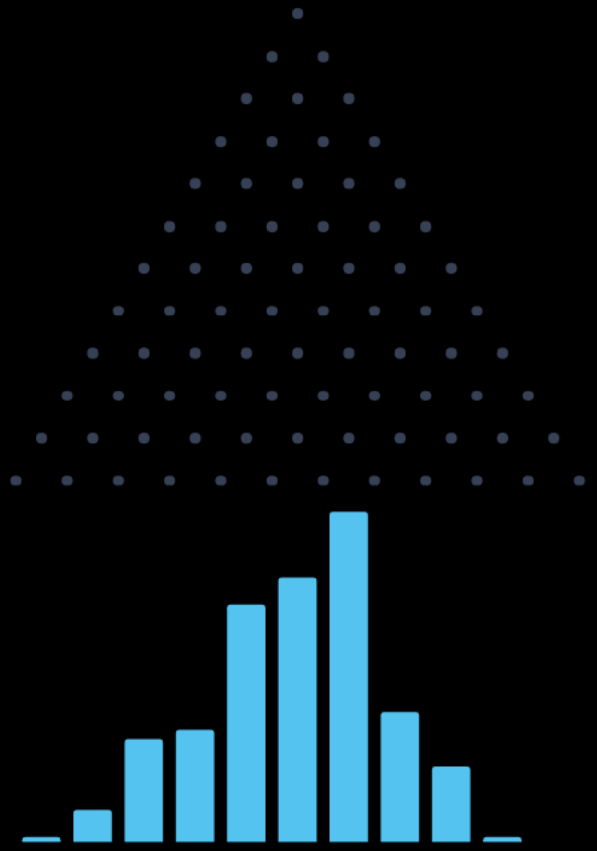


Random Variables

Learning goals:

1. A random variable is a number assigned to each possible outcome of something random.

2. It has a probability mass function (PMF for short) that determines the probability of each outcome. Really, this is like a histogram (or set of bars) that shows how likely each value is.

3. Expectation (also known as the mean) is the balance point of the PMF. Variance is how far values "wander" from the mean.



One coin. One person faking it.

Sequence 1

TTHHTHTTHTTTHTTTHTTTHTTHTHHTHHTHTHHTTTHHTHTHTTHTHH

TTHTHHTHTTTHHTTHHTTHHHTHHTHTTHTHTTHHTHHHTTHTHTTTHH

TTHTHTHTHTHTTHTHTHHHTTHTHTHHTHHHTHTHTTHTTHHTHTHTHT

THHTTHTHTTHHHTHTHTHTTHTTHHTTHTHHTHHHTTHHTHTTHTHTHT

HTHTHTHHHTHTHTHTHHTHHTHTHTTHTTTHHTHTTTHTHHTHHHHTT

HHTHTHTHTHHHTTHHTHTTTHTHHTHTHTHHTHTTHTTHTHHTHTHTT

Sequence 2

HTHHHTHTTHHTTTTTTTTHHHTTTHHTTTHHTTHHHTTHTHTTTTTTH

THTTTTHHHHTHTHTTHTTTHTTHTTTTHTHHTHHHTTTTTHHHHTHHH

TTTTHTHTTHHHHTHHHHHHHHTTHHTHHTHHHHHHHTTHTHTTTHHTTT

THTHHTTHTTHTHTHTTHHHHTTHTTTHTHTHHTTTTHTTTTTHHTHTH

HHHTTTTHTHHHTHHTHTHTHTHHHTHTTHHHTHHHHHTHHHTHTTTHH

HTTTHHTHTTHHTHHHTHTTHTTHTTTHHTHTHTTTTHTHTHTTHTHTHT

Which one is the real coin?

From CS109 Problem Set 2, “What is Random?”: 300 real flips, and 300 typed by a person trying to look random.

Random variables



Random variable

A number assigned to each possible outcome of something random.

X = number of heads in 300 coin flips

$X = 148$ an event: true or false

$\mathrm{P}(X = 148)$ a probability: a number between 0 and 1



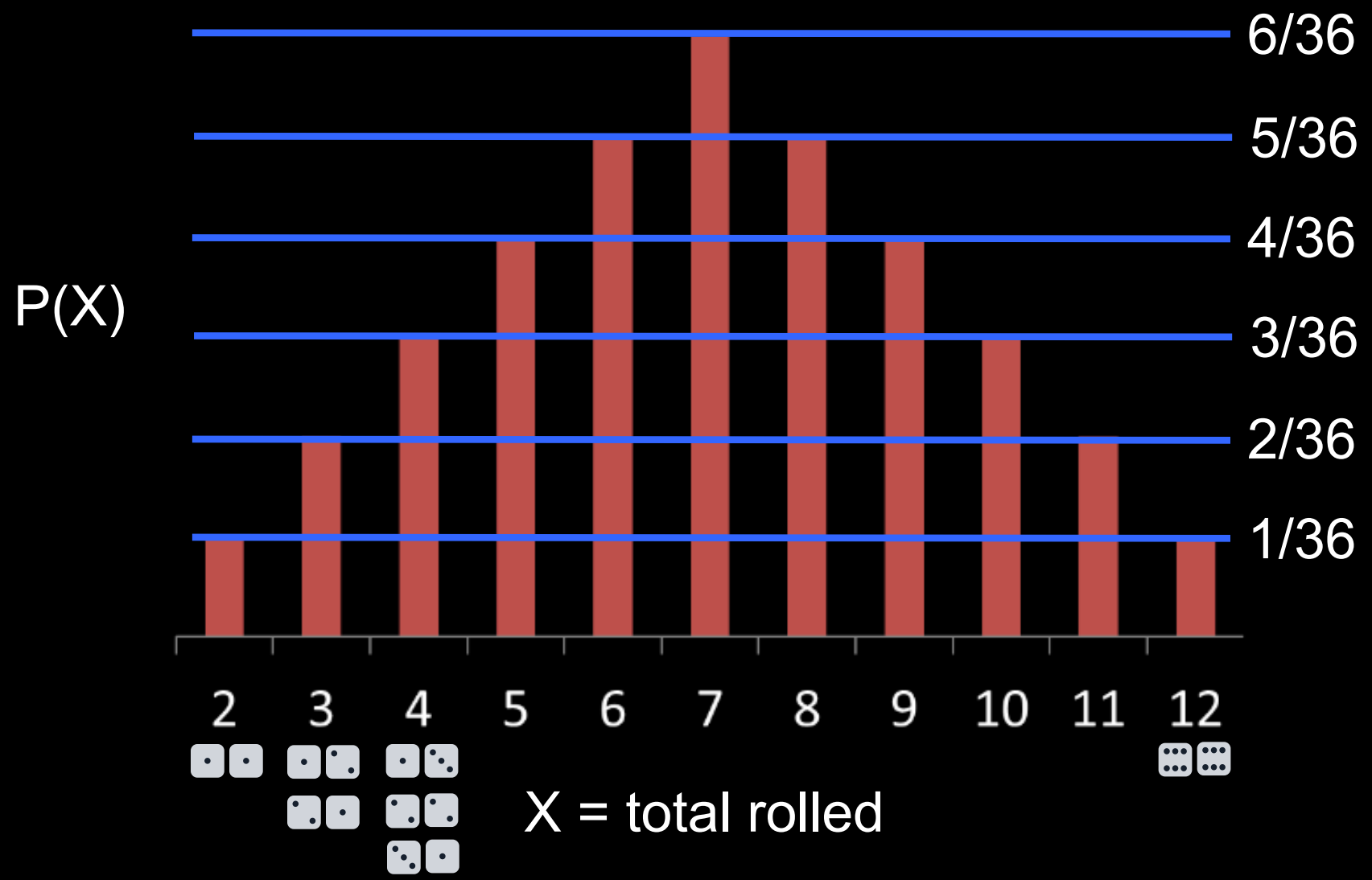


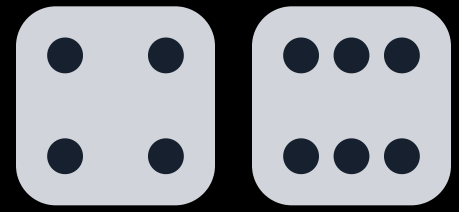
The most important thing to know about a random variable: its probability mass function (PMF).

It's like a histogram where there is one bar per value, as tall as its probabilty. The bars all add up to 1.

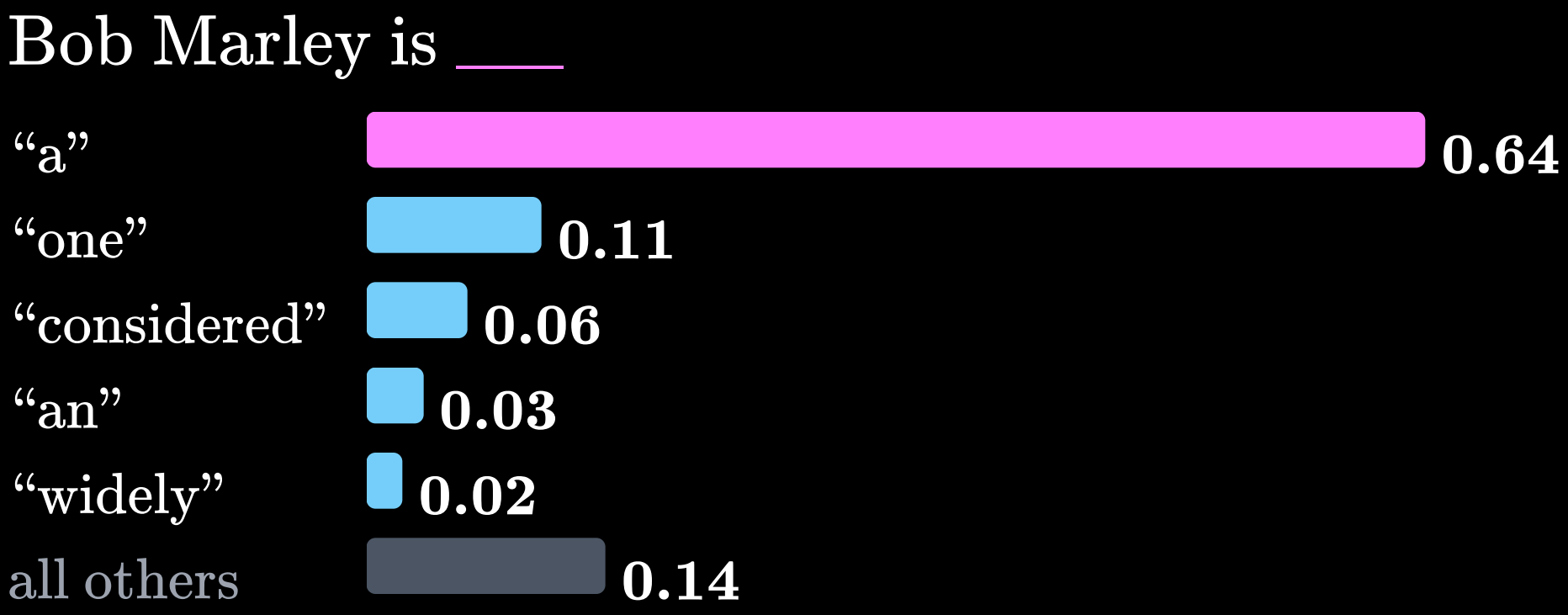


PMF for Rolling Two 6-Sided Dice





An AI's next word has PMF too



Gemini 2.5 Flash's next-word chances after "Bob Marley is". The bars add up to 1.

Expectation

the balance point



Expectation

Expectation (also called the Mean),
is the balance point of the PMF, defined as:

$$E[X] = \sum_{X=x} x\, P(X = x)$$

Example: Expectation of rolling a 6-sided die 

$$E[X] = 1(\frac{1}{6}) + 2(\frac{1}{6}) + 3(\frac{1}{6}) + 4(\frac{1}{6}) + 5(\frac{1}{6}) + 6(\frac{1}{6}) = \frac{21}{6} = 3.5$$



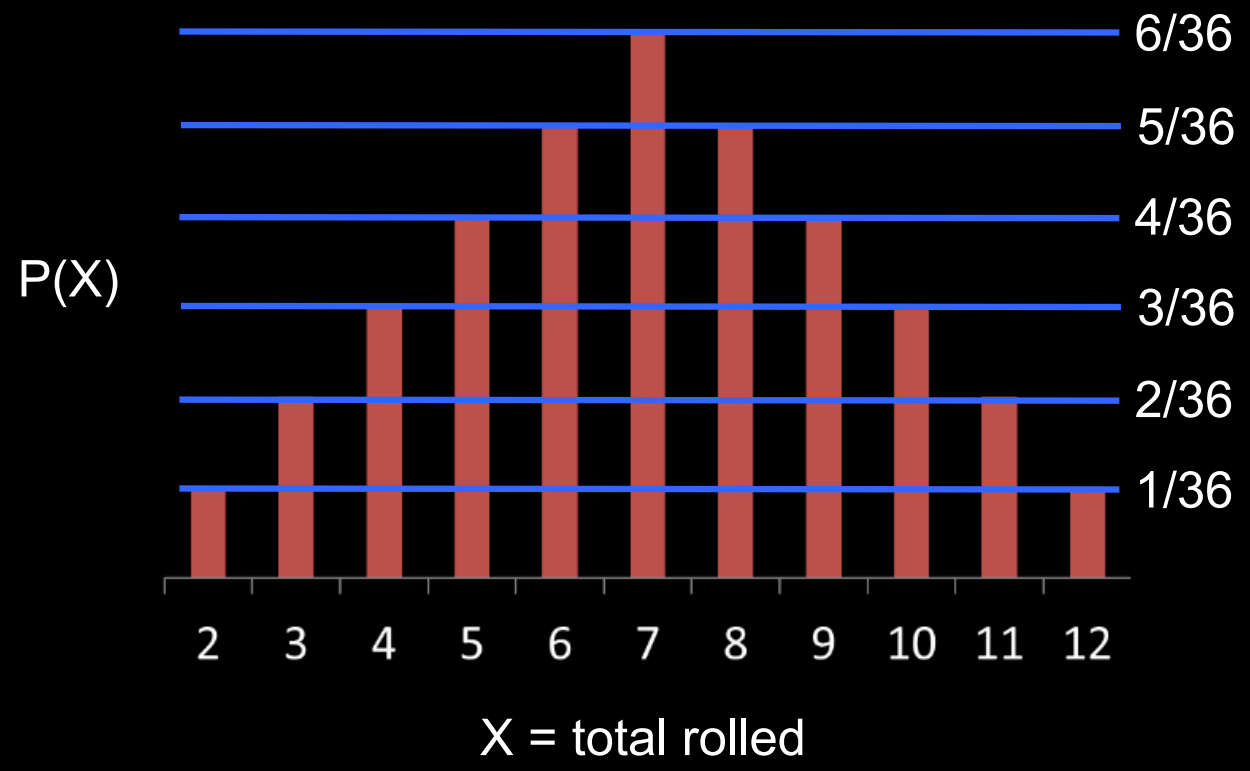
Roll a die



Expectation: the long-run average.

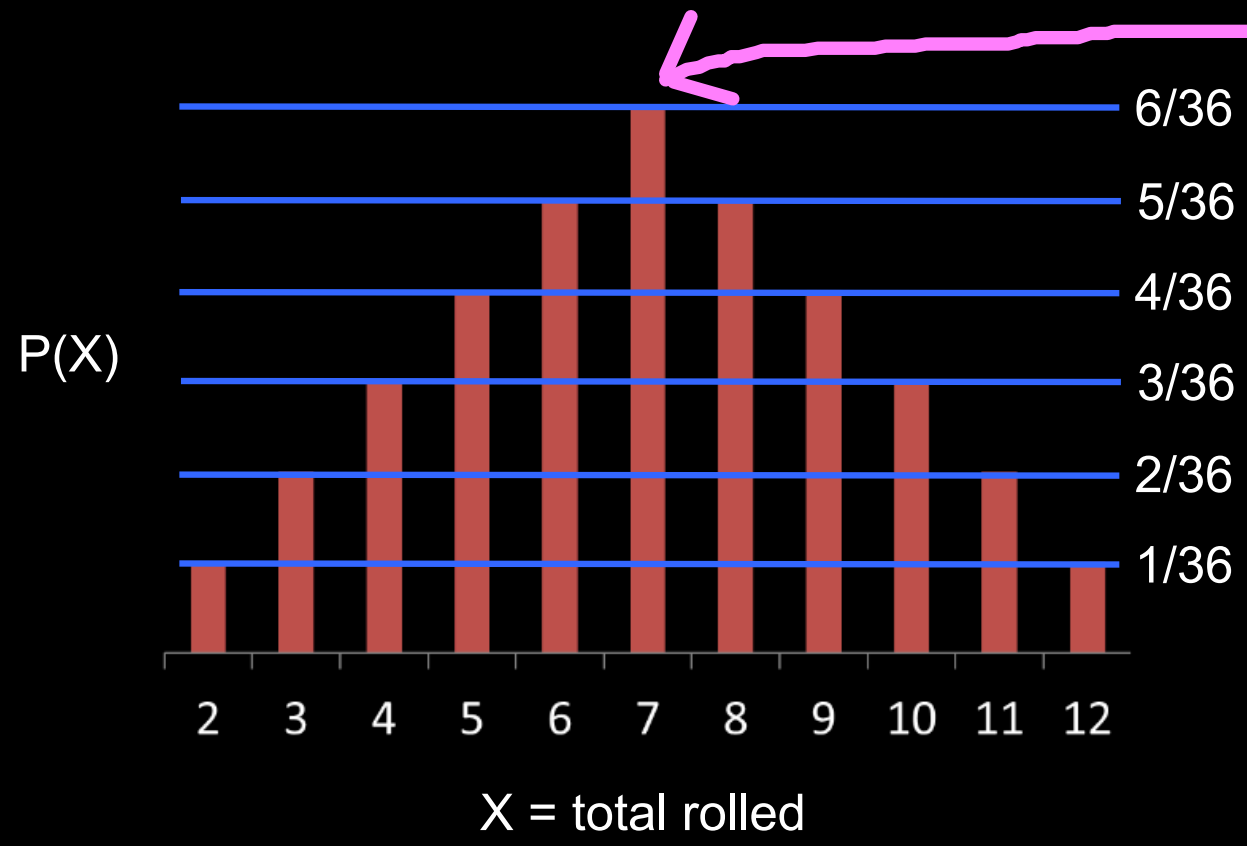
No roll ever gives 3.5.

Recall PMF for Rolling Two 6-Sided Dice



$$E[X] = 2(\frac{1}{36}) + 3(\frac{2}{36}) + 4(\frac{3}{36}) + 5(\frac{4}{36}) + ... + 12(\frac{1}{36}) = \frac{252}{36} = 7$$

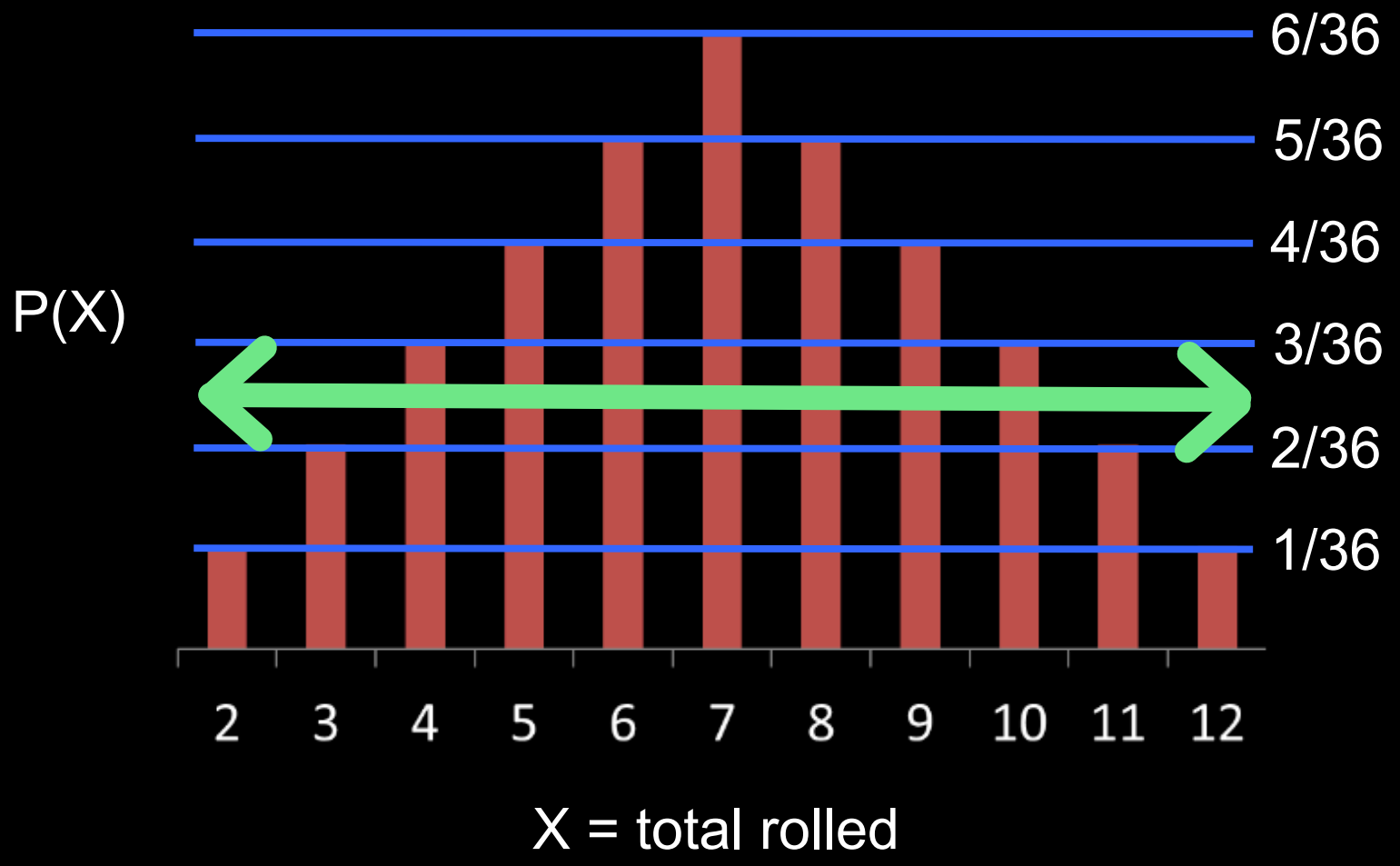
Recall PMF for Rolling Two 6-Sided Dice



Balance point of PMF

$$E[X] = 2(\frac{1}{36}) + 3(\frac{2}{36}) + 4(\frac{3}{36}) + 5(\frac{4}{36}) + \ldots + 12(\frac{1}{36}) = \frac{252}{36} = 7$$

Recall PMF for Rolling Two 6-Sided Dice



Variance is amount of spread in values around the mean

Formula for variance in course reader

Common Types of Random Variables (Distributions)



Bernoulli Random Variable

A random variable with two values (0 or 1)

$$\mathrm{P}(X = 1) = p \qquad \mathrm{P}(X = 0) = 1 - p$$

X is a Bernoulli random variable: X~Ber(p)

$E[X] = 1(p) + 0(1 - p) = p$

Examples:

- Coin flip
- Random binary digit (bit)



Jacob Bernoulli

Jacob Bernoulli (1654-1705), also known as “James”, was a Swiss mathematician



Namesake for Bernoulli random variable

Fun Fact: He is my academic great^{11}-grandfather

Binomial Random Variable

Represents n independent trials of Bernoulli random variable

X is number of "successes" (1's) in n trials

X is a Binomial random variable: X~Bin(n, p)

Formula for PMF in course reader

$$E[X] = np$$

Examples:

- Number of heads in n coin flips
- Number of 1's in randomly generated string of n bits

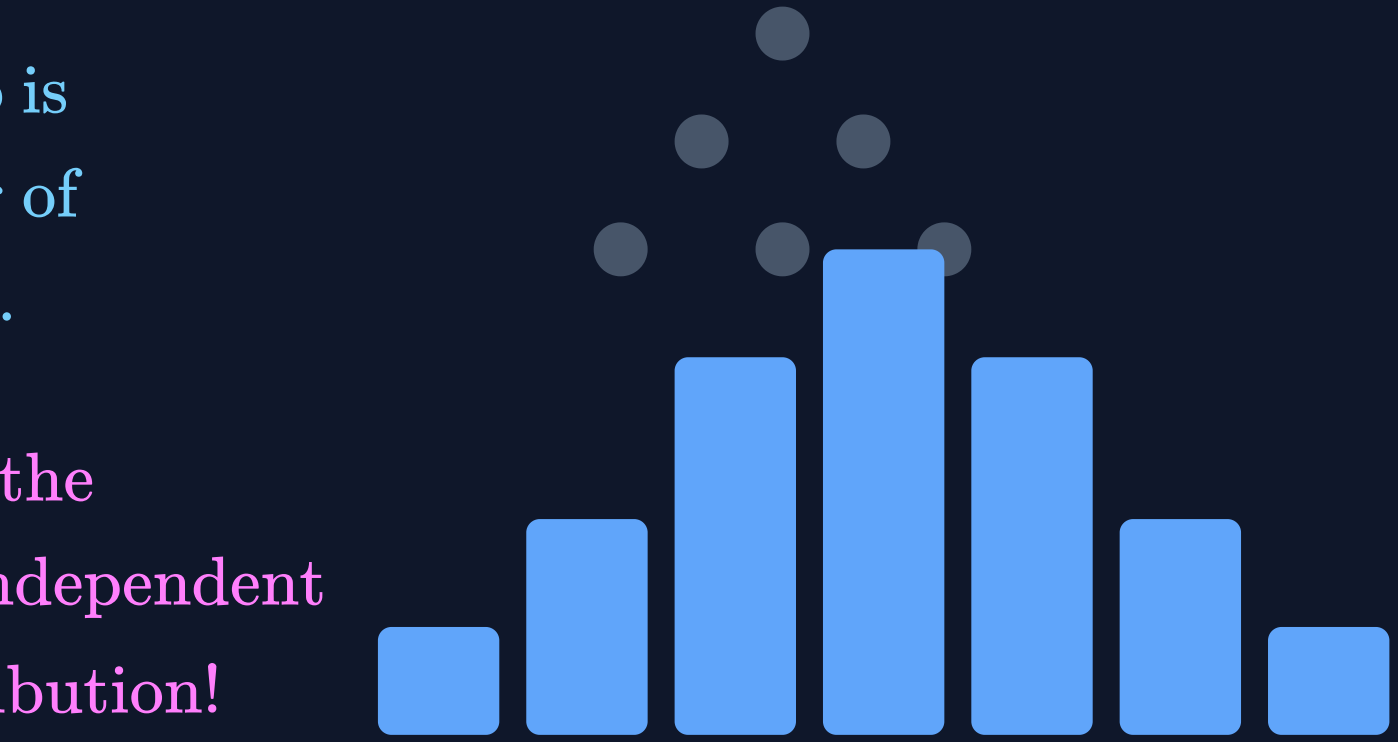


One peg, one coin flip

At each peg: ball goes left or right, 50/50.

Which bucket ball ends up is determined by the number of times it goes right at a peg.

The number of “rights” is the number of successes in n independent trials: it’s a binomial distribution!



Poisson Random Variable

Approximates a Binomial when n is large and p is small

X takes on values 0, 1, 2, 3, ...

X is a Poisson random variable: $X \sim \text{Poi}(\lambda)$

$E[X] = \lambda \approx np$

Generally good approximation when:

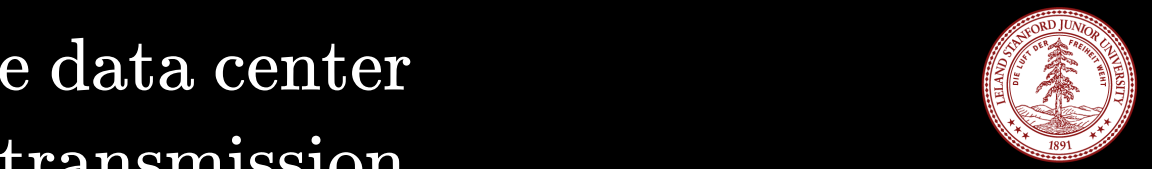
$n > 20$ and $p < 0.05$

$n > 100$ and $p < 0.1$

Examples:

Number of component failures in large data center

Number of bits corrupted in message transmission



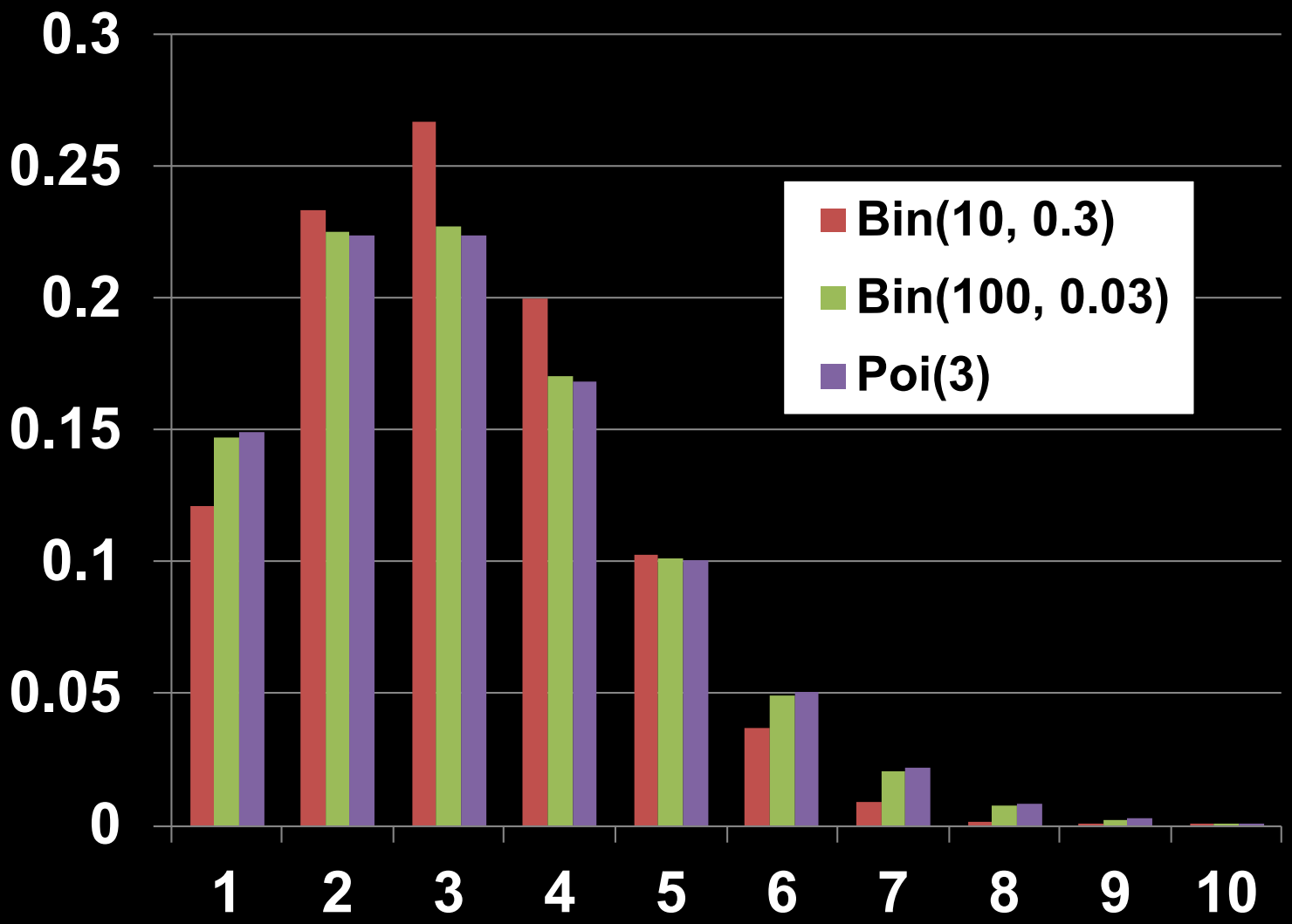
Simeon-Denis Poisson

Simeon-Denis Poisson (1781-1840) was a prolific French mathematician

He reportedly said “Life is good for only two things: discovering mathematics and teaching mathematics”



PMFs for Binomials versus Poisson



Note that expectation $= 3$ for all three distributions

Poi(3) and Bin(100, 0.03) match very well because n is large and p is small

Chocolate Chip Cookies

- Making *many* cookies with *lots* of chocolate chips
- Equal sized clumps of batter for each cookie
  - Probability p that a particular chocolate chip is in a certain cookie is small ($p = 1/\#$ cookies)
  - There are a large number n of chocolate chips
  - X = number of chocolate chips in a cookie



$$\text{X} \sim \text{Poi}(\lambda), \text{ where } \lambda = \frac{\#\text{ chocolate chips}}{\#\text{ cookies}}$$

Can use Poisson distribution for *spatial* modeling
(Number of events per fixed amount of space)

Counting Flashes in Radioactive Decay

1910: count the flashes of radioactive decay
in each 7.5-second window.



Slice up window in many slivers, each with a small chance of a flash. (p is small)

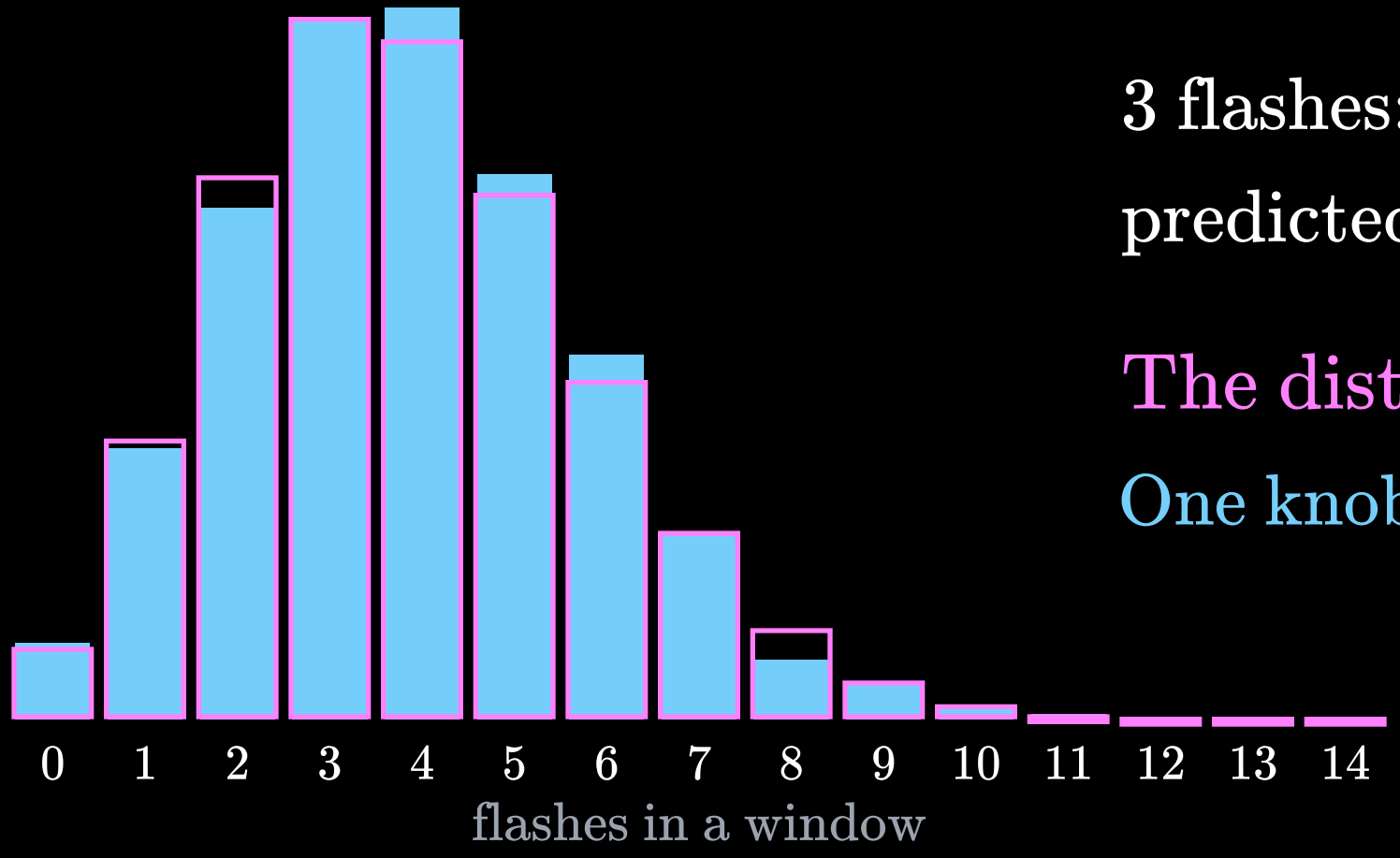
There are many slivers. (n is large)

Total number of flashes within each 7.5 second window:
approximated by Poisson random variable.

Can use Poisson distribution for *temporal* modeling
(Number of events per fixed interval of time)

E. Rutherford and H. Geiger, "The probability variations in the distribution of α particles," Philosophical Magazine, 1910.

Number of Flashes in 2,608 Windows



3 flashes: 525

predicted: 525

The distribution is a Poisson.

One knob: the mean = $\lambda = 3.87$

Rutherford and Geiger, 1910: 10,097 flashes in 2,608 windows.

Recall: One coin. One person faking it.

Sequence 1

TTHHTHTTHTTTHTTTHTTTHTTHTHHTHHTHTHHTTTHHTHTHTTTHTHH

TTHTHHTHTTTHHTTHHTTHHHTHHTHTTHTHTTHHTHHHTTHTHTTTHH

TTHTHTHTHTHTTHTHTHHHTTHTHTHHTHHHTHTHTTHTTHHTHTHTHT

THHTTHTHTTHHHTHTHTHTTHTTHHTTHTHHTHHHTTHHTHTTHTHTHT

HTHTHTHHHTHTHTHTHHTHHTHTHTTHTTTHHTHTTTHTHHTHHHHTTT

HHTHTHTHTHHHTTHHTHTTTHTHHTHTHTHHTHTTHTTHTHHTHTHTTT

Sequence 2

HTHHHTHTTHHTTTTTTTHHHTTTHHTTTTHHTTHHHTTHTHTTTTTTH

THTTTTHHHHTHTHTTHTTTHTTHTTTTHTHHTHHHHTTTTTHHHHTHHH

TTTTHTHTTHHHHTHHHHHHHHTTHHTHHTHHHHHHHTTHTHTTTHHTTT

THTHHTTHTTHTHTHTTHHHHHTTHTTTHTHTHHTTTTHTTTTTHHTHTH

HHHTTTTHTHHHTHHTHTHTHTHHHTHTTHHHTHHHHHTHHHTHTTTHH

HTTTHHTHTTHHTHHHTHTTHTTHTTTHHTHTHTTTTHTHTHTTHTHTHT

Which one is the real coin?

$$300 \times \frac{1}{2} = 150$$

On average: number flips x the probablity of heads

Sequence 1: 148 heads. Sequence 2: 148 heads.

Count of heads can't tell them apart.

From CS109 Problem Set 2, "What is Random?": 300 real flips, and 300 typed by a person trying to look random.

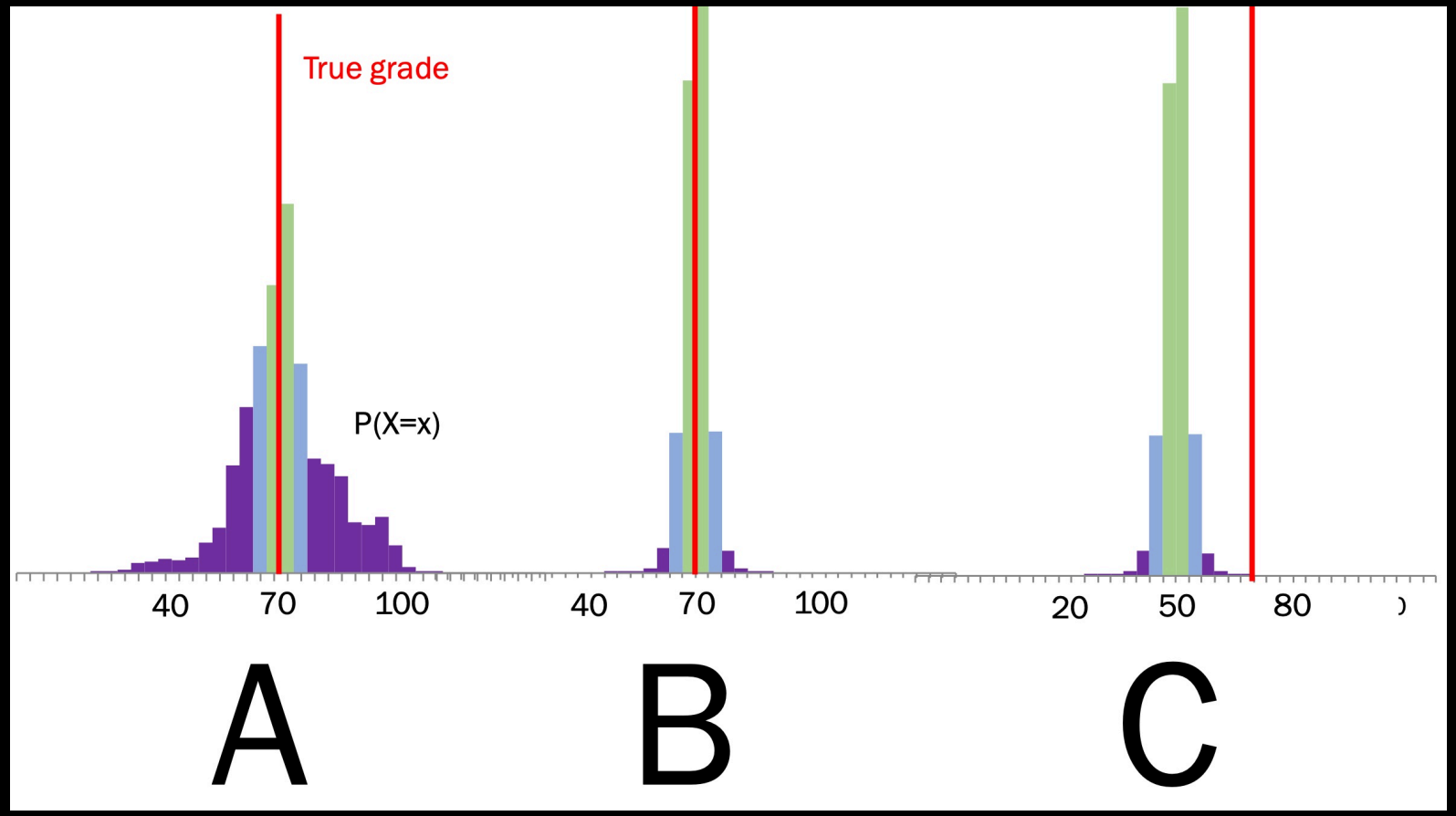
Variance

Intuitively, the “spread” -- how far values wander from the mean

Often use “standard deviation” to refer to the “spread”



Three kinds of peer graders



A, B: balance on the true grade.

B wanders less: less spread.

C: little spread, wrong balance point.

Grades from a massive online course. C. Piech et al., "Tuned Models of Peer Assessment in MOOCs," EDM 2013.



Did an AI write this?





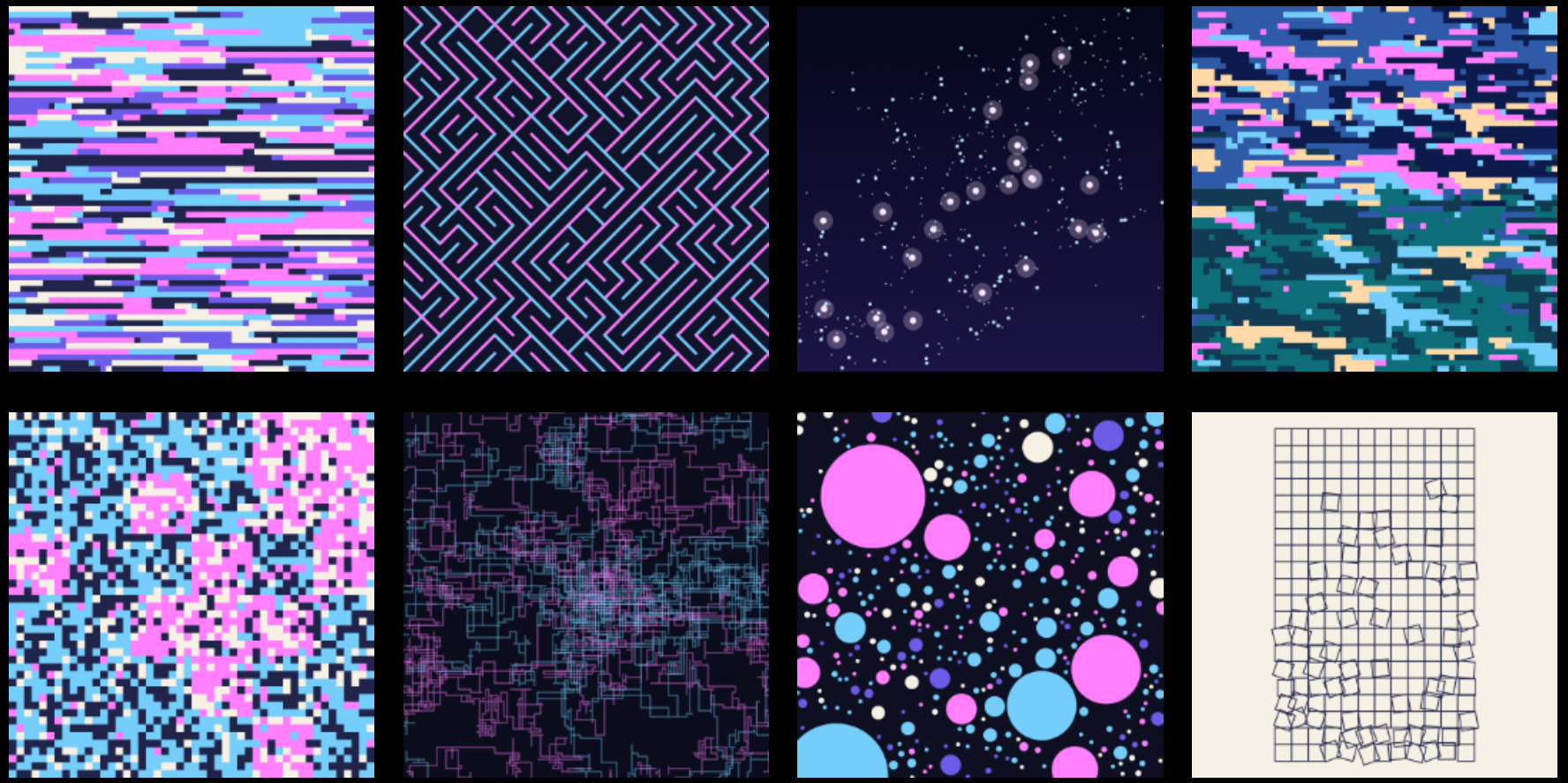
J. Kirchenbauer et al., “A Watermark for Large Language Models,” ICML 2023, Figure 1. Their green list is drawn blue.



Count, then ask: how many spreads away?

The same move caught the person faking coin flips and the AI.
Answer B has 8 pink pieces and still gets caught.

Next: stories with knobs



Each one is a story with a few knobs. Your next build: Art Detective